Questão 1

```
None
def get_employee_status(employee):
    return "Senior" if is_experienced(employee) else "Junior"



def is_experienced(employee):
    return employee["years_of_experience"] >= 5
```

Questão 2

```
None
def calculate_invoice(invoice):
    total = invoice["quantity"] * invoice["unit_price"]
    tax = total * 0.15
    final_amount = total + tax

    print("Invoice total:", total)
    print("Tax:", tax)
    print("Final amount:", final_amount)

    return final_amount
```

Questão 3

```
None
def get_shipping_cost(order):
```

```
    return calculate_delivery_fee(order)



def calculate_delivery_fee(order):
    return 25 if order["weight"] > 10 else 10
```

Questão 4

```
None
def register_user(user):
    print("Validating user information...")

    if not user["name"]:
        print("Invalid name")
        return False

    if not user["email"]:
        print("Invalid email")
        return False

    print("Saving user...")
    print("Sending confirmation email...")

    return True
```

Questão 5

None

```
def calculate_salary(employee):

    base_salary = employee["salary"]

    bonus = base_salary * 0.10

    deductions = base_salary * 0.05

    final_salary = base_salary + bonus - deductions



    return final_salary
```

Questão 1

None

```
def process_order(order):
    customer = order["customer"]
    items = order["items"]

    subtotal = 0

    for item in items:
        subtotal += item["quantity"] * item["unit_price"]

    if customer["category"] == "Premium":
        discount = subtotal * 0.15
    elif customer["category"] == "Regular":
        discount = subtotal * 0.05
    else:
        discount = 0

    total = subtotal - discount

    print("Order processed successfully")
    print("Customer:", customer["name"])
    print("Items:", len(items))
```

```
    print("Subtotal:", subtotal)
    print("Discount:", discount)
    print("Total:", total)

    return total
```

Questão 2

None

```
def authenticate_user(user, password):
    if user is None:
        return False

    if user["status"] != "active":
        return False

    if not check_password(password, user["password"]):
        return False

    return True


def check_password(input_password, stored_password):
    return input_password == stored_password
```

Questão 3

None

```
def generate_library_report(books):
    available_books = []
    borrowed_books = []

    for book in books:
        if book["available"]:
            available_books.append(book)
        else:
            borrowed_books.append(book)
```

```
    print("LIBRARY REPORT")
    print("----------------------")
    print("Total books:", len(books))
    print("Available books:", len(available_books))
    print("Borrowed books:", len(borrowed_books))

    print("\nAvailable titles:")

    for book in available_books:
        print("-", book["title"], "|", book["author"])

    return len(available_books), len(borrowed_books)
```

Questão 4

None

```
def update_inventory(product, quantity_sold):
    current_stock = product["stock"]

    if quantity_sold > current_stock:
        print("Insufficient stock")
        return False

    product["stock"] = current_stock - quantity_sold

    if product["stock"] == 0:
        product["status"] = "Out of stock"
    elif product["stock"] <= 5:
        product["status"] = "Low stock"
    else:
        product["status"] = "Available"

    print("Inventory updated")
    print("Product:", product["name"])
    print("Remaining stock:", product["stock"])
    print("Status:", product["status"])

    return True
```

Questão 5

None

```
def calculate_employee_payment(employee):
    hours_worked = employee["hours_worked"]
    hourly_rate = employee["hourly_rate"]

    regular_hours = min(hours_worked, 40)
    overtime_hours = max(hours_worked - 40, 0)

    regular_payment = regular_hours * hourly_rate
    overtime_payment = overtime_hours * hourly_rate * 1.5

    gross_payment = regular_payment + overtime_payment

    tax = gross_payment * 0.10
    net_payment = gross_payment - tax

    print("PAYMENT SUMMARY")
    print("----------------------")
    print("Employee:", employee["name"])
    print("Regular payment:", regular_payment)
    print("Overtime payment:", overtime_payment)
    print("Gross payment:", gross_payment)
    print("Tax:", tax)
    print("Net payment:", net_payment)

    return net_payment
```